

Case Study – Phase II (Design)

Architectural Modelling (CO2)

Group No

: 10

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1. Problem Title: Vehicle Rental Management System

STRUCTURAL DESIGN

2. Class Diagram

The class diagram below illustrates all seven classes identified in Phase I, along with their attributes, methods, and inter-class relationships using standard UML notation. Associations, aggregations, and dependency arrows represent structural relationships between entities.

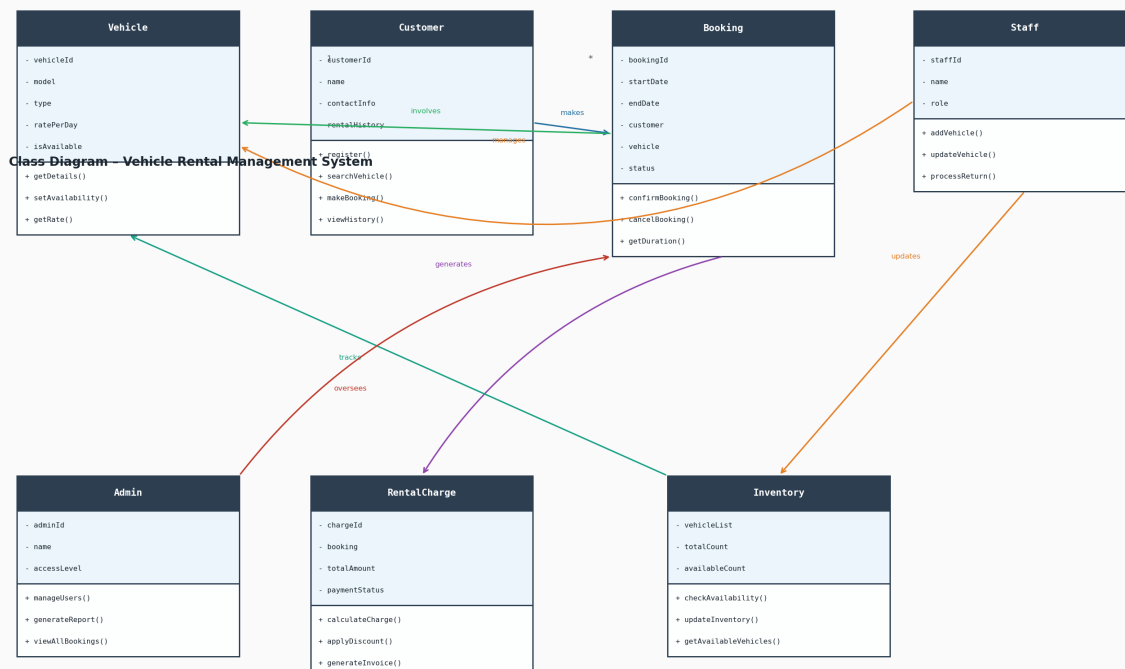


Figure 1: UML Class Diagram – Vehicle Rental Management System

BEHAVIOURAL DESIGN

3. Use-Case Diagram

a. List of Actors

#	Actor	Role
1	Customer	Registers, searches vehicles, makes/cancels bookings, views history
2	Staff	Adds/updates vehicles, processes vehicle returns
3	Admin	Manages users, views all bookings, generates reports
4	System	Automatically tracks availability, calculates charges

b. List of Use-Cases

UC#	Use Case	Description
UC1	Register / Login	Customer creates an account or logs into the system.
UC2	Search Vehicle	Customer searches for available vehicles by type and date.
UC3	Make Booking	Customer selects a vehicle and confirms a rental booking.
UC4	View Booking History	Customer or Admin views past and current rental records.
UC5	Cancel Booking	Customer cancels an existing booking before the rental starts.
UC6	Calculate Rental Charge	System computes total charge based on duration and vehicle rate.
UC7	Add / Update Vehicle	Staff adds new vehicles or updates existing vehicle information.
UC8	Process Vehicle Return	Staff records the return of a rented vehicle and updates status.
UC9	Manage Users	Admin manages customer and staff accounts.
UC10	Generate Report	Admin generates system-wide rental and revenue reports.
UC11	Track Availability	System automatically tracks and updates fleet availability in real time.

c. Use-Case Diagram (UML)

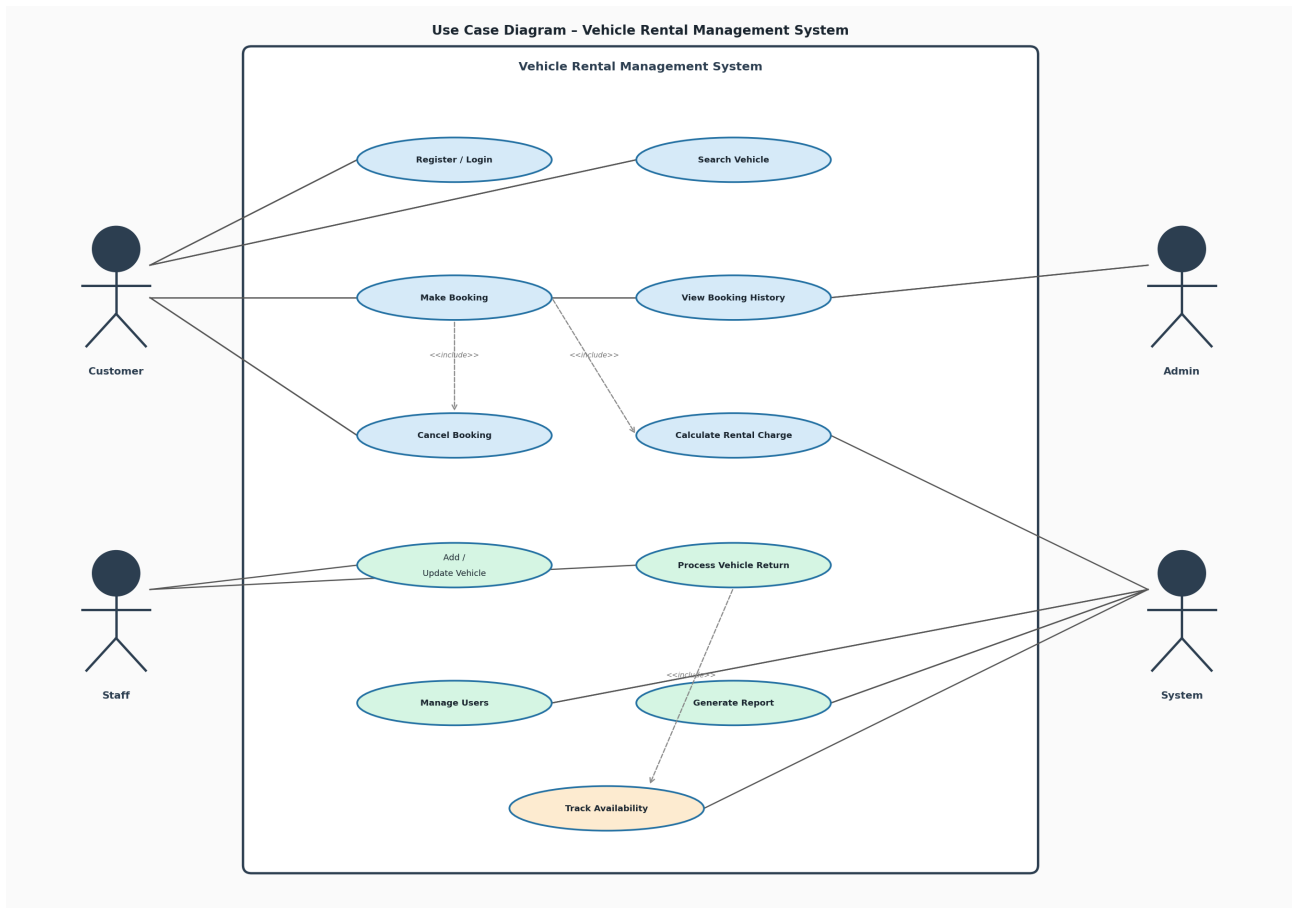


Figure 2: UML Use-Case Diagram – Vehicle Rental Management System

4. Sequence Diagrams (UML)

a. Scenario: Customer Makes a Booking

A customer searches for an available vehicle by specifying the vehicle type and rental dates. The system queries the Inventory to find available vehicles and fetches their details. Once the customer selects a vehicle and confirms the booking, a Booking object is created, the Inventory is updated to mark the vehicle as unavailable, and RentalCharge calculates the total cost and issues an invoice. A confirmation is then displayed to the customer.

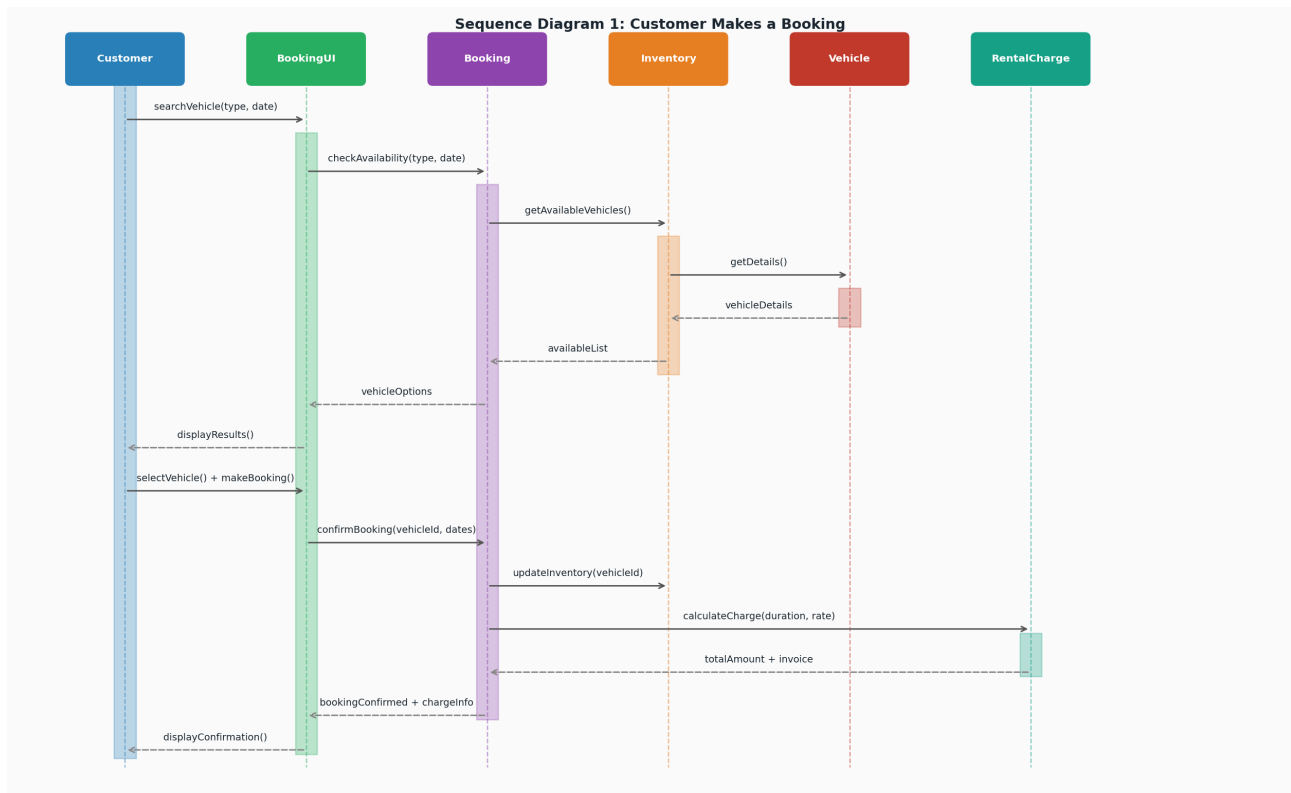


Figure 3: Sequence Diagram – Customer Makes a Booking

b. Scenario: Staff Processes Vehicle Return

When a customer returns a rented vehicle, the Staff initiates the return process by providing the booking ID. The system retrieves the booking details and computes the actual rental duration. The Vehicle availability status is updated to available, and the Inventory is refreshed to reflect the change. RentalCharge then calculates the final charge based on actual duration, generates an invoice, and the Staff receives a return summary to hand over to the customer.

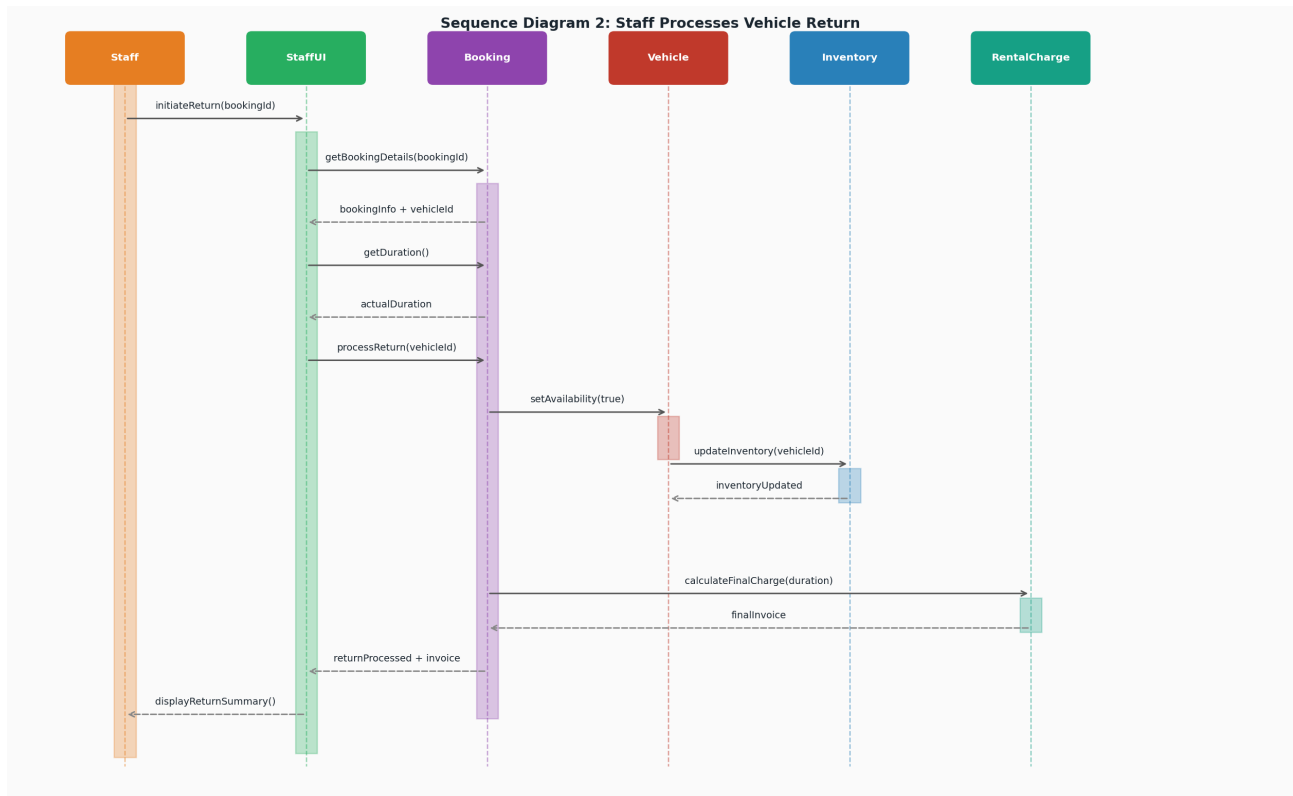


Figure 4: Sequence Diagram – Staff Processes Vehicle Return